

Finally, in some applications, an op-amp that can supply significant amounts of current, say 100 mA, may be needed. To get such a large current, we may need to increase the widths of the output MOSFETs in the push-pull amplifier to, for example, 10,000 and 5,000 (for the PMOS and NMOS devices, respectively). The problem with this is that the quiescent current that flows in these devices, using the op-amp configuration in Fig. 24.44 as an example, is 1 mA. To reduce this current, we can increase the lengths of the floating current sources used in the folded-cascode section. This increases their gate-source voltage drops and moves the gates of the push-pull MOSFETs towards the power supply rails (shutting them off). Care must be exercised when doing this because shutting off the output MOSFETs can result in moving g_{m2} towards zero, causing, as indicated in Eq. (24.24), f_2 to move down in frequency, thus making the op-amp unstable.

24.4 Gain-Enhancement

While using minimum channel lengths ($L = 1$) results in the fastest op-amps, the open-loop gain, $A_{OLD C}$, is considerably reduced. Using gain-enhancement (GE) techniques (regulating the drain node in a cascode structure, see Fig. 20.40), we boost the low-frequency gain by increasing the cascode output resistance. However, the gain-bandwidth product, f_{un} , is still $g_m/2\pi C_c$. (We get the speed by using a smaller compensation capacitor and/or a larger diff-amp transconductance.) Note, in Fig. 24.47 for example, that our gain-bandwidth product went up by the increase in g_m with the reduction in the channel length. Reviewing Eqs. (24.30)–(24.34), we see that if $A_{OLD C}$ goes up, f_{3dB} goes down (and f_{un} is a constant).

In the following discussion, we only concern ourselves with two-stage op-amps in which we are using GE to compensate for the low gain caused by our minimum channel lengths. While GE can be used in an OTA, the fact that the speed of the OTA is limited by the capacitance it is driving keeps them from being as fast as the two-stage op-amps we've discussed (unless, of course, the load capacitance is small).

As seen in Fig. 20.40, we need an additional amplifier to help regulate the drain of the cascode device. Consider the two differential amplifiers (OTAs) seen in Fig. 24.50. The source-followers on the inputs of the diff-amps are used to allow for the amplification of signals near ground (for the P amplifier) or V_{DD} (for the N amplifier). The gains of the diff-amps are

$$A_P = g_{mp} \cdot (r_{on} || r_{op}) \text{ and } A_N = g_{mn} \cdot (r_{on} || r_{op}) \quad (24.55)$$

To simplify the equations, we'll assume

$$A_{GE} = A_P = A_N \quad (24.56)$$

The low-frequency gain of the op-amp in Fig. 24.44 is

$$A_{OLD C} = \overbrace{g_{mn} \cdot (R_{ocasn} || R_{ocas p})}^{\text{diff-amp gain}} \cdot \overbrace{(g_{mp} + g_{mn}) \cdot (r_{on} || r_{op})}^{\text{push-pull output stage}} \quad (24.57)$$

Using Eq. (20.67), we can write the open-loop gain *with gain-enhancement* as

$$A_{OLD C, GE} = A_{OLD C} \cdot A_{GE} \quad (24.58)$$

The open-loop gain is increased by the gain of the added amplifier. Figure 24.51 shows the op-amp of Fig. 24.44 with GE.

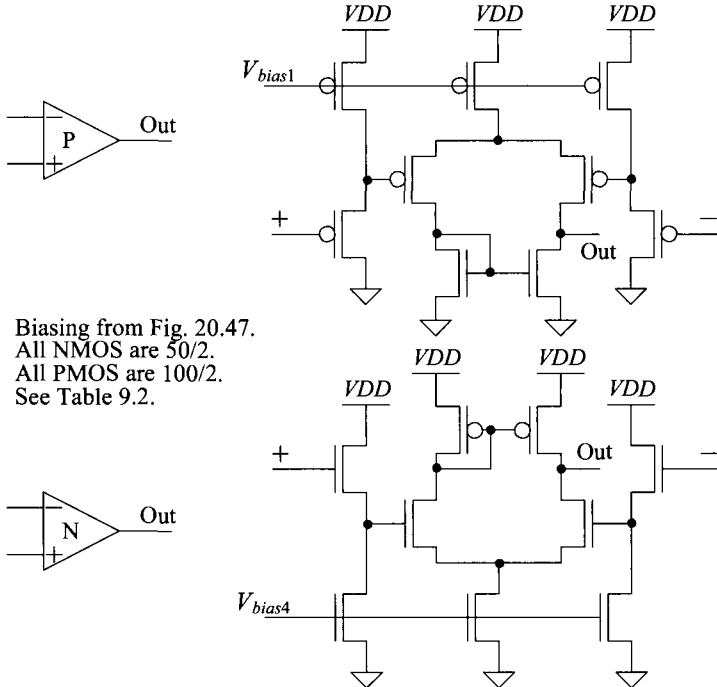


Figure 24.50 Diff-amps with source-follower level shifters for use in GE.

The simulation results in Fig. 24.47 were generated using the op-amp of Fig. 24.44 with the lengths reduced from 2 to 1. The low-frequency, open-loop gain was 53 dB. In Fig. 24.51 we added GE and used the resulting op-amp to generate the simulation results seen in Fig. 24.52 (with the GE compensation capacitors, C_{cGE} , set to zero). The low-frequency, open-loop gain is now 74 dB. The step response is seen as well. The current pulled from VDD is now 60 μA higher than the op-amp without GE. By replacing the diff-amps in the N and P amplifiers in Fig. 24.50 with folded-cascode OTAs (like the one seen in Fig. 24.42), we can get a greater increase in open loop gain. However, the stability of the regulated-drain feedback loop becomes even more important.

Bandwidth of the Added GE Amplifiers

Let's model the frequency response of the added amplifiers in Fig. 24.50 by

$$A_{GE}(f) = \frac{A_{GEDC}}{1 + j \frac{f}{f_{3dBGE}}} \quad (24.59)$$

Using this equation with Eqs. (24.30) and (24.58), we can write

$$A_{OLGE}(f) = \frac{A_{OLDLC}}{1 + j \frac{f}{f_{3dB}}} \cdot \frac{\overbrace{A_{GEDC}}^{A_{GE}(f)}}{1 + j \frac{f}{f_{3dBGE}}} \quad (24.60)$$

Biasing using Fig. 20.47.
 Unlabeled NMOS are 50/1.
 Unlabeled PMOS are 100/1.

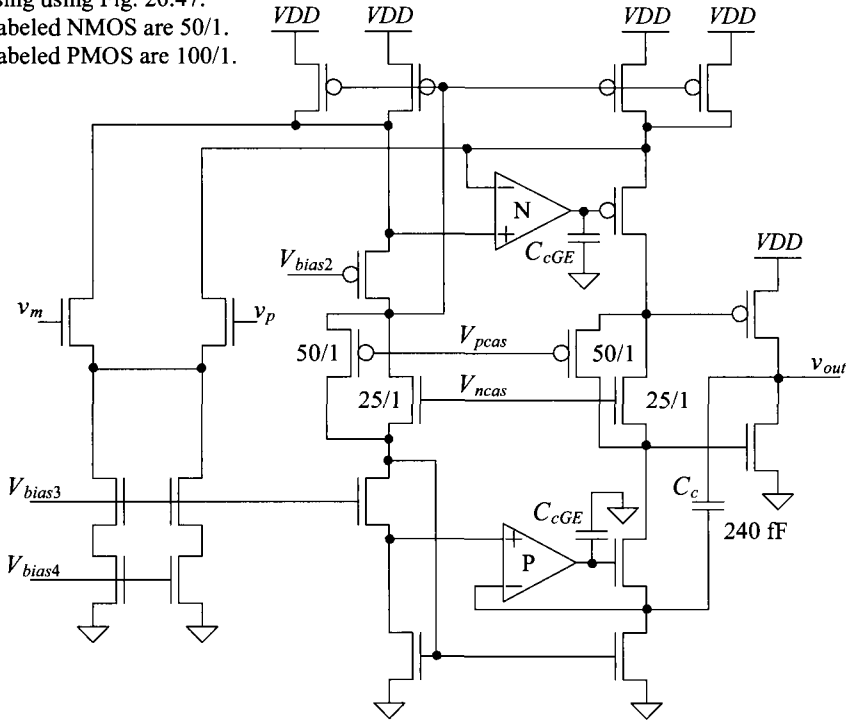


Figure 24.51 Folded-cascode op-amp with class AB output buffer and gain-enhancement.

However, the pole ($f_1 = f_{3dB}$) on the output of the folded-cascode OTA can be written, see Eqs. (20.67) and (24.47), with GE as

$$f_{3dB} = \frac{1}{2\pi(R_{ocasn} || R_{ocasp}) \cdot A_{GE}(f) \cdot C_c} \quad (24.61)$$

assuming the current through C_c is larger than the other currents, the output of the OTA may have to charge. If we substitute Eq. (24.61) into Eq. (24.60) and look at frequencies much larger than f_{3dB} (which from Fig. 24.52 is only a 100 kHz or so), we see that the frequency response of the added GE amplifiers cancels out of $A_{OLGE}(f)$. What this means is that the bandwidth of the added amplifiers doesn't need to be wide. As long as the bandwidths are larger than the f_{3dB} of the op-amp, the GE works as desired. Also, notice that when we substitute Eq. (24.61) into Eq. (24.60), the resulting transfer function shows a *doublet* (a pole and zero at the same frequency). If the pole and zero aren't at exactly the same frequency, then the settling time of the amplifier can be affected by the change in phase shift (noting that in Eq. (20.67) we are approximating the output resistance of the cascode structure). The output resistance of the cascode structures is decreasing with increasing frequency, while, at the same time, the decrease in the output resistance results in an increase in the circuit bandwidth (causing the gain-bandwidth product to be a constant).

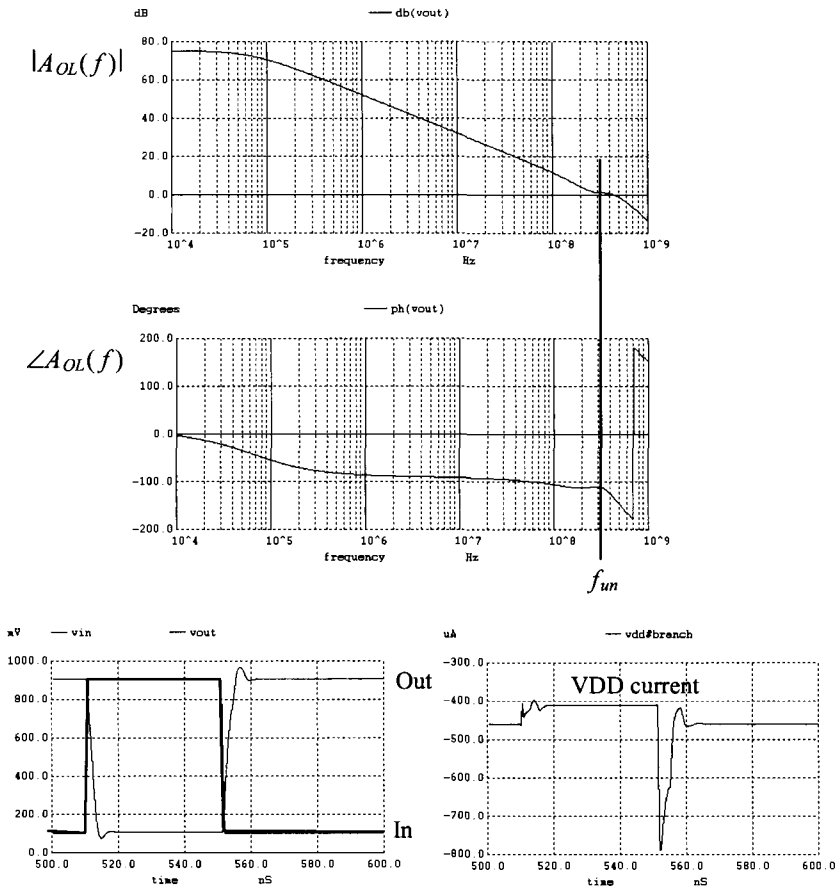


Figure 24.52 Operation of the op-amp in Fig. 24.51 driving a 1-pF load.

Compensating the Added GE Amplifiers

When we generated the simulations in Fig. 24.52, we didn't have compensation capacitors on the output of the added GE OTAs (N and P); that is, C_{cGE} were zero. The problem with this is that the resulting feedback loops aren't necessarily stable. As seen in Fig. 24.52, the step response shows some overshoot (indicating a low phase margin). Also, the frequency response isn't as monotonic as we would like (-20 dB/dec) around f_{un} . As with any feedback structure, compensation is important. *So we need to add capacitors to the outputs of the GE OTAs.* Figure 24.53 shows the response of the op-amp in Fig. 24.51 when C_{cGE} s are set to 240 fF (the GE OTAs then have f_{un} of 100 MHz, as calculated using Eq. (24.44)). The frequency response is more monotonic and the step response doesn't show overshoot and ringing. The gain-bandwidth product of the op-amp is 400 MHz. Note, it may be useful to regenerate Fig. 24.53, using larger values of C_{cGE} to prove to oneself that indeed the bandwidth of the added GE amplifiers isn't critical. Also, it may be useful to replace the basic diff-amps with higher-gain OTAs to show that the low-frequency gain of the op-amp does indeed increase as Eq. (24.58) shows.

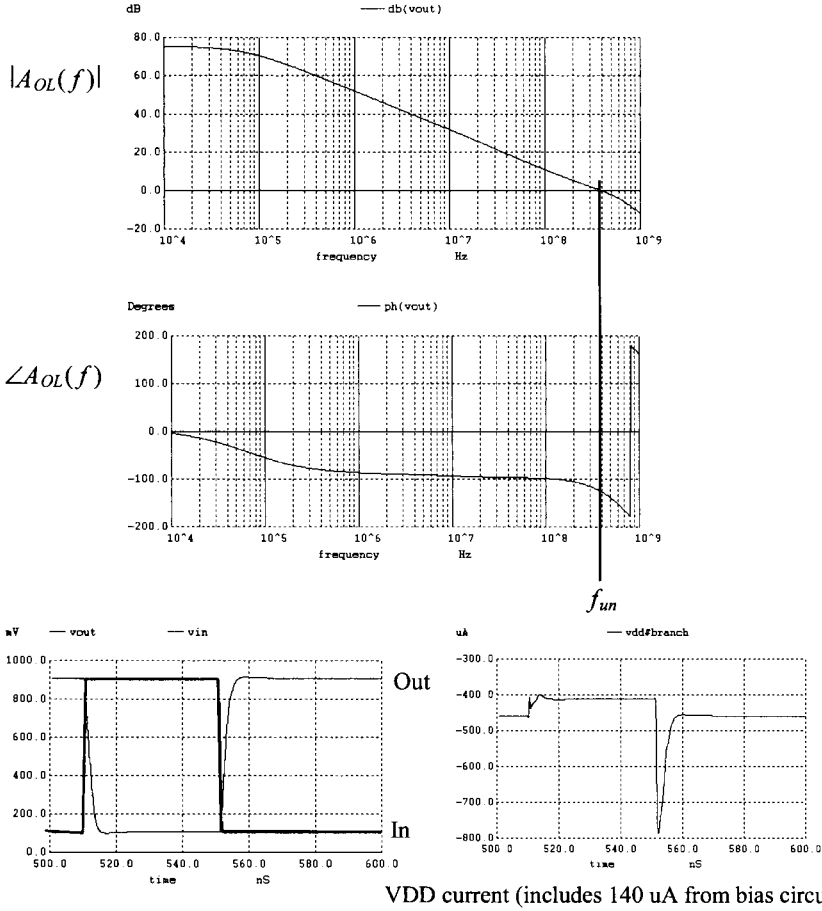


Figure 24.53 Operation of the op-amp in Fig. 24.51 driving a 1-pF load with GE compensation capacitors of 240 fF.

24.5 Some Examples and Discussions

A Voltage Regulator

One of the applications of an op-amp is in regulating a voltage on-chip. Figure 24.54 shows the basic topology. A voltage reference is used with the op-amp to generate a regulated voltage, V_{REG} . If the voltage reference is stable with temperature, the fact that the V_{REG} is a function of a ratio of resistors (so process or temperature changes in the resistance value don't affect the ratio) and the variation in the op-amp's open loop gain is desensitized using feedback makes the regulated voltage stable with process and temperature changes. The ideal (meaning that the op-amp has infinite open-loop gain) regulated voltage is

$$V_{REG} = V_{REF} \cdot \left(1 + \frac{R_A}{R_B} \right) \quad (24.62)$$

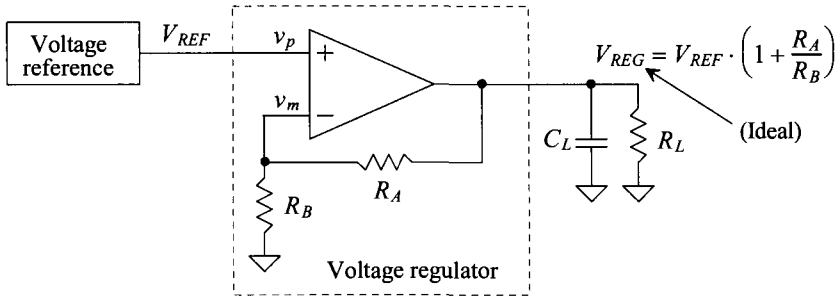


Figure 24.54 Schematic of a voltage regulator.

If the op-amp's open-loop gain is finite, then we can write

$$V_{REG} = A_{OL} \cdot (v_p - v_m) \quad (24.63)$$

and

$$v_m = V_{REG} \cdot \frac{R_B}{R_A + R_B} \text{ and } v_p = V_{REF} \quad (24.64)$$

Solving for the actual regulated voltage, gives

$$V_{REG} = V_{REF} \cdot \frac{1}{\frac{1}{A_{OL}} + \frac{R_B}{R_A + R_B}} \quad (24.65)$$

noting that when A_{OL} is infinite (or very large) this equation simplifies to Eq. (24.62). Equation (24.65) can be very revealing. Let's say that V_{REF} is 500 mV, R_B is an open, and R_A is a short (the op-amp is a voltage follower). Ideally, V_{REG} is also 500 mV. If we use a diff-amp with an open-loop gain of 20, then the actual regulated voltage is

$$V_{REG} = 0.5 \cdot \frac{1}{1.05} = 476 \text{ mV} \quad (24.66)$$

This simple example illustrates *why open-loop gain is important*. When op-amps are used in data converters (discussed in Chs. 28-30), they must amplify signals to within mV, which requires op-amp open-loop gains in the thousands.

Figure 24.55 shows a basic regulator topology using the two-stage op-amp. The regulator only sources current and so we've made the PMOS device, on the output, very large. One concern with using the large PMOS device is the op-amp offset (which can be considerable, see Fig. 24.4, for example). Another concern occurs when simulating. In some simulators both NRD and NRS (number of squares of implant area in the source or drain of a MOSFET) defaults to one. If the sheet resistance, r_{sh} , is 50 ohms, then the resistance in series with the MOSFET is 100 ohms (limiting the current the MOSFET can supply). For large MOSFETs, it is generally a good idea to set NRD and NRS to zero. An example of a (large) SPICE MOSFET statement is

Mbig vout vn1 VDD VDD PMOS L=1 W=10000 NRD=0 NRS=0

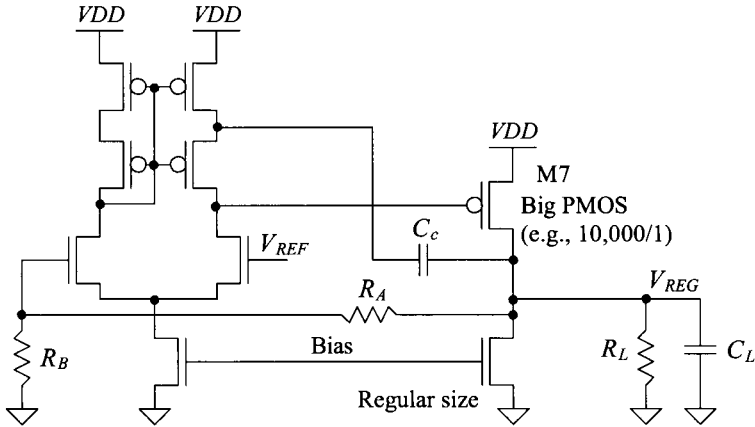


Figure 24.55 Schematic of a voltage regulator.

The point of making M7 big was to ensure that the regulator could supply a significant amount of current. However, notice that the gate of M7 cannot swing any lower (assuming that all MOSFETs remain in saturation) than

$$V_{G7,\min} = V_{REF} - V_{THN} \quad (24.67)$$

For example, if V_{REF} is 500 mV and V_{THN} is 250 mV, then the minimum voltage on the gate of M7 is 250 mV (25% of VDD). Ideally, we would like the gate of M7 to be able to swing to ground, fully turning it on. Towards this goal, consider the modified voltage regulator seen in Fig. 24.56. In this figure, we've used the OTA from Fig. 24.33 for the first stage of the op-amp. Now, when M5 turns on and M4 shuts off, the gate of M7 can get yanked down towards ground, allowing M7 to turn fully on.

In any practical regulator, a bypass capacitor (a big capacitor placed from V_{REG} to ground) is used to supply charge to the load for fast current transients. The required bandwidth of the regulator is reduced since the bypass capacitor (part of C_L in Figs. 24.55 and 24.56) takes care of the fast transients. It is desirable to have a large C_L for “filtering” the load's fast current needs. However, from our discussions earlier, this (large load capacitance) makes compensating the op-amp more challenging. In all of our compensation schemes, the pole associated with the output of the first stage was at a frequency much lower than the pole associated with the output stage (this was one of the reasons we used pole-splitting). To isolate the load from the second-stage gain, we might insert an NMOS source-follower (SF) between the load and the output of the second stage. The problem with this is that we won't be able to turn the SF on enough to supply significant amounts of current. Next, we might try replacing the common-source amplifier (M7) with a class AB stage (like the topology seen in Fig. 24.29). Here, again, the output won't be able to supply as much current as what we get with a full VDD across M7's V_{SG} . (Note that one of the reasons for using a voltage regulator is to hold V_{REG} constant even if VDD approaches V_{REG} .) If we must use a large load capacitor to supply charge for the fast variations in the load current and if the topology in Fig. 24.56 is used, then *we must try to compensate the op-amp with the load capacitance*.

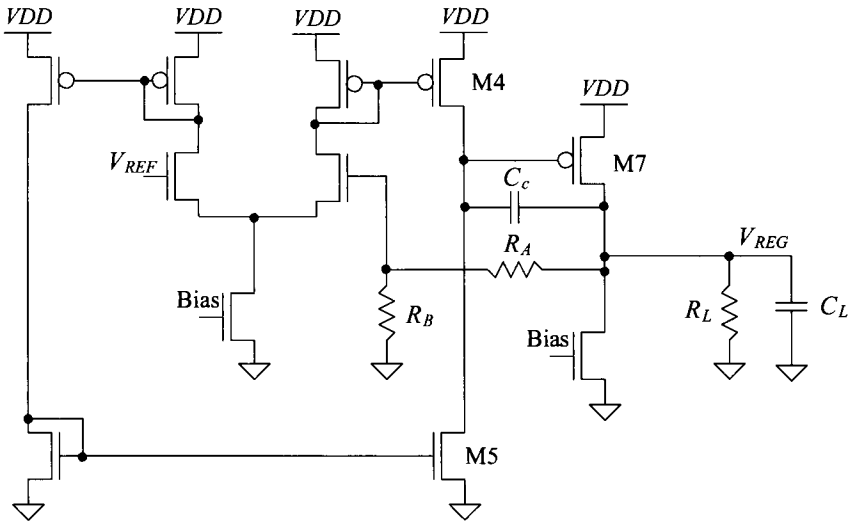


Figure 24.56 Schematic of a voltage regulator where the gate of M7 can be pulled all the way to ground.

If we are compensating the op-amp (the voltage regulator) using the load capacitance, then we've got to make sure that there is a minimum value of C_L . Further, increasing the load capacitance should make the regulator more stable, slowing the response of the op-amp. The reduction in the speed of the op-amp is offset by the fact that a larger load capacitance can supply more charge to a fast current transient before the op-amp must respond.

Why use a compensation capacitor, C_c , if we are compensating with the load capacitance? The compensation capacitor pushes the pole associated with the output node to a higher frequency (exactly what we don't want to do). We still use the capacitor (but with a smaller value) for large-signal reasons. If V_{REG} drops suddenly, the decrease in voltage is fed back directly to the gate of M7 through C_c . This turns M7 (quickly) on and allows it to pull V_{REG} back up (bypassing the slower feedback action of the op-amp). Note then that C_c is *not used* for compensating the op-amp (and actually makes the regulator more unstable). Further note that increasing the gain of the op-amp by making the first-stage gain larger (by, say, using a cascode diff-amp to increase the output resistance of the diff-amp, R_1) also hurts the regulator's stability. (By using a larger R_1 , the pole associated with the output of the first stage is moved lower in frequency.)

To estimate the location of the output pole, assuming that it is the dominant pole in the system, let's calculate the closed-loop output resistance of the op-amp $R_{out,CL}$ (the open-loop output resistance is R_2). We know that the low-frequency, open-loop gain of the op-amp is

$$A_{OLDC} = \frac{v_{out}}{v_p - v_m} = g_{m1}R_1g_{m2}R_2 \quad (24.68)$$

If we replace the load in Fig. 24.54 with a test voltage, v_t , and note that V_{REF} ($= v_p = 0$) is AC ground, then with the help of Fig. 24.57 we can write

$$i_t = \frac{v_t}{R_2} + \frac{v_t}{R_A + R_B} + g_{m1}R_1g_{m2} \cdot \frac{v_t}{A_{CL}} \quad (24.69)$$

If we assume that the current through the feedback path ($R_A + R_B$) is small, then the output resistance of the regulator is

$$R_{out,CL} \approx \frac{v_t}{i_t} = \frac{1}{\left(\frac{1}{R_2} + \frac{g_{m1}R_1g_{m2}}{A_{CL}}\right)} \approx \frac{A_{CL}}{A_{OLDC}} \cdot R_2 \quad (24.70)$$

Our output pole is located at

$$f_2 = \frac{1}{2\pi R_{out,CL} \cdot C_L} = \frac{A_{OLDC}}{2\pi R_2 C_L \cdot A_{CL}} \quad (24.71)$$

For this pole to compensate the op-amp, we want to use a low value of open-loop gain, A_{OLDC} . Again, this means that we don't want a large value of resistance on node 1 (the output of the diff-amp). The problem with this is that the lower the open-loop gain, the more difficult it is to regulate the output voltage, as indicated in Eq. (24.66).

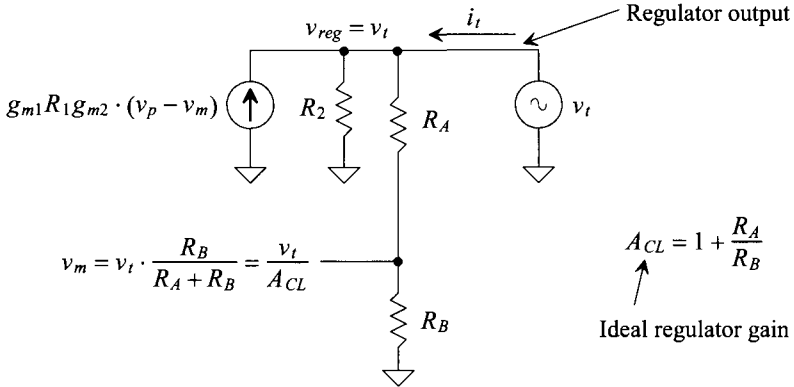


Figure 24.57 Determining the closed-loop output resistance of a two-stage op-amp.

To get an idea for the minimum size load capacitance required for a stable op-amp, let's use the parameters from Table 9.2. To begin, note that the more current the regulator supplies to a load (the more current flowing through M7), the smaller the value of R_2 . This (decreasing R_2) means that the regulator becomes more likely to be unstable (f_2 increases) as it supplies more current. However, at the same time, the open-loop gain, A_{OLDC} , drops and the ability of the op-amp to regulate the output declines.

Let's assume M7's r_o and g_m are the values given in Table 9.2. Because M7 is so wide, this means (using these values for r_o and g_m) that it is conducting very little current. The value of f_1 can be estimated (assuming the $C_{gs7} \gg C_c$ and the width of M7 is 10,000 or 100 times larger than the widths of the PMOS in Table 9.2) as

$$f_1 \approx \frac{1}{2\pi R_1 C_{gs7}} = \frac{1}{2\pi \cdot (167k || 333k) \cdot 100 \cdot 8.34f} = 1.7 \text{ MHz} \quad (24.72)$$

The DC open-loop gain of the op-amp is calculated using Eq. (24.68) as 277 (no load). If we use the op-amp in the follower configuration where R_b is infinite and R_A is zero (A_{CL} is 1), then (knowing $R_2 = 167k || 333k = 111k$)

$$f_2 = \frac{A_{OLDC}}{2\pi R_2 C_L \cdot A_{CL}} = \frac{277}{2\pi(111k) \cdot C_L} \quad (24.73)$$

Knowing we want $f_2 \ll f_1$, $A_{OLDC} = 277$, and $f_1 = 1.7$ MHz, we can set the unity-gain frequency to 1 MHz (making sure that it is less than f_1). If we want a -20 dB/decade roll-off above f_2 (now the low-frequency pole) then, as seen in Eq. (24.33), we can write

$$f_{un} = A_{OLDC} \cdot f_2 = \frac{A_{OLDC}^2}{2\pi R_2 C_{Lmin} \cdot A_{CL}} = 1 \text{ MHz} \quad (24.74)$$

Solving for the minimum value of C_{Lmin} using these equations gives 110 nF. Clearly, this value is too large if we want the regulator to be completely on-chip. Rewriting Eq. (24.74)

$$C_{Lmin} = \frac{(g_{m1} R_1 g_{m2})^2 R_2}{2\pi \cdot f_{un} \cdot A_{CL}} \quad (24.75)$$

Note that linearly increasing the current through M7 causes R_2 to decrease linearly (assuming that M7's output resistance dominates R_2) and g_{m2}^2 to increase linearly (see Eqs. (9.6) and (9.22)). The result is that C_{Lmin} doesn't change. However, notice what happens if we include R_L in the calculations. If $R_L \ll r_{o7}$, then we can write

$$\frac{C_{Lmin}}{R_{Lmax}} = \frac{(g_{m1} R_1 g_{m2})^2}{2\pi \cdot f_{un} \cdot A_{CL}} \quad (24.76)$$

If we select $C_{Lmin} = 1000$ pF, then $R_{Lmax} = 1$ k Ω . We use the variable R_{Lmax} to indicate that this is the maximum value of load resistance possible for a stable regulator (we want the regulator to always supply at least V_{REG}/R_{Lmax} current). If the regulator's output voltage is 500 mV, it must be supplying at least 500 μ A of current. This resistor keeps the open-loop gain low when the regulator is supplying small amounts of current. We might think, after looking at Eq. (24.73), that by supplying more current (R_L or R_2 decreasing further), the location of f_2 increases and the regulator moves towards instability. However, for further decreases in the load resistance, the current supplied by M7 increases and thus so does its g_m . What happens is this: we get the canceling of effects. As the current supplied to the output increases linearly with a linear decrease in R_L , the value of g_m^2 increases linearly and the effects cancel.

A final comment: because this design is different from the previous op-amp designs, thorough characterization using simulations is needed. Steps in the load current as well as a wide range of load capacitances and resistances should be simulated to ensure that the regulator remains stable under all possible VDD s, temperatures, and process shifts.

Bad Output Stage Design

In this section we'll discuss some problems with biasing in Class AB output stages. To begin, consider the op-amp design seen in Fig. 24.58. This topology is a diff-amp driving an inverter. The problem with this design is that the current flowing in the push-pull output stage (an inverter) isn't set by a bias circuit. Because of this, the current may be

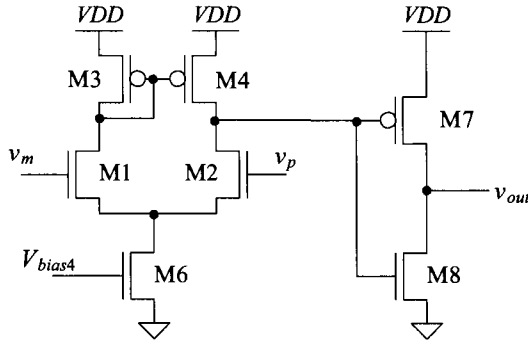


Figure 24.58 Bad op-amp design.

significant and vary greatly with both temperature and process shifts. Further, notice that when the inputs to the op-amp are equal, the drain of M4 is at the same potential as the drain of M3 (the currents in M3 and M4 are equal). This, as we've already discussed, can be used to set the bias current in M7 (we treat M7, for biasing purposes, as if its gate were tied to the gates of M3 and M4). If we call the gate potential of M7 (and thus the gate potential of M8) $VDD - V_{SG}$, then for M8 to be in saturation

$$V_{out} \geq VDD - V_{SG} - V_{THN} \quad (24.77)$$

If VDD is 1 V, $V_{SG} = 350$ mV, and $V_{THN} = 250$ mV, then V_{out} must be greater than 400 mV for M8 to be saturated. What this means is that for wide output swings, M8 will move into triode and the push-pull output amplifier gain will drop (killing the overall gain of the op-amp). What we need to do is drop the gate potential of M8 down so that its drain can swing to a lower voltage without it trioding. Towards this goal, consider adding a source-follower level shifter to the output of the op-amp, as seen in Fig. 24.59. Now the output voltage in the amplifier can swing lower without M8 trioding. However, we still aren't controlling the current in the output stage. By not controlling this current, as we've already discussed, we produce an op-amp with poor systematic, input-referred offset voltage. If, for example, we have an input-referred offset of 10 mV and the diff-amp gain is 20, then the change in the output voltage on the gate of M7 is 200 mV. This can cause

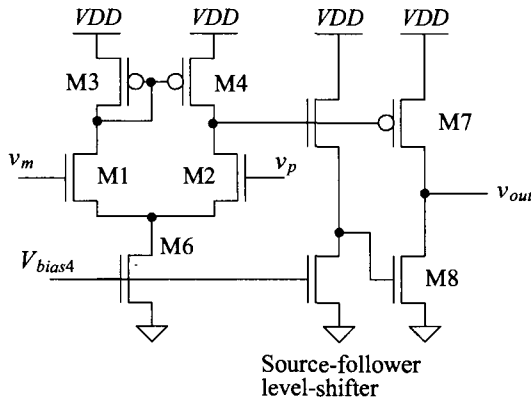


Figure 24.59 Another bad op-amp design.